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MTH642 - Fluid Mechanics Finalterm latest quiz spring 2026

Section 1: Bernoulli's Equation and Applications

1. According to Bernoulli's principle for an ideal fluid, as the speed of the fluid increases, the pressure:
   a) Increases
   b) **Decreases**
   c) Remains constant
   d) Becomes zero

Correct Answer: b

2. The mathematical proportionality relating pressure $P$to velocity $v$in a moving fluid is:
   a) $P \propto v$
   b) $P \propto v^2$
   c) $P \propto \frac{1}{v}$
   d) $P \propto \frac{1}{v^2}$

Correct Answer: c

3. In a constricted artery, the flow speed increases, causing the internal pressure to:
   a) Increase
   b) Stay the same
   c) **Drop, potentially causing collapse**
   d) Double

Correct Answer: c

4. Which of the following is **NOT** an application of Bernoulli's equation?
   a) Flight of an airplane
   b) Perfume spray

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c) Venturi effect
d) **Melting point of ice**

Correct Answer: d

5. Bernoulli's equation is valid for:
   a) Viscous fluids
   b) **Ideal fluids**
   c) Turbulent flow only
   d) All real fluids

Correct Answer: b

6. The SI unit of pressure (Pa) is equivalent to:
   a) J·m
   b) J/m²
   c) **J/m³**
   d) N·m

Correct Answer: c

7. The term $\frac{1}{2}\rho v^2$ represents:
   a) Potential energy density
   b) **Kinetic energy per unit volume**
   c) Static pressure
   d) Internal energy

Correct Answer: b

8. The term ρgh represents:
   a) Kinetic energy
   b) **Gravitational potential energy per unit volume**
   c) Viscous drag
   d) Elastic energy

Correct Answer: b

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9. Bernoulli's Principle explains that lift in birds and planes is mainly generated:
   a) Above the cockpit
   b) In the tail
   c) **Under the wings**
   d) In the engine

Correct Answer: c

10. On the top of an airfoil, air typically has:
   a) **Higher speed and lower pressure**
   b) Lower speed and higher pressure
   c) No airflow
   d) High pressure and high speed

Correct Answer: a

11. On the bottom of an airfoil, air typically has:
   a) Higher speed and lower pressure
   b) **Lower speed and higher pressure**
   c) Vacuum
   d) Atmospheric pressure only

Correct Answer: b

12. A car spoiler is best described as:
   a) A type of engine
   b) **A reversed airfoil**
   c) A fuel stabilizer
   d) A braking device

Correct Answer: b

13. In a viscous fluid flowing through a pipe, pressure gradually:
   a) Increases
   b) **Decreases**

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c) Fluctuates
d) Remains constant

Correct Answer: b

14. Torricelli's formula is used to find velocity of liquid:
a) In a venturi meter
b) **Flowing out of a small hole**
c) In a large artery
d) Falling in vacuum

Correct Answer: b

15. Torricelli's formula shows the exit velocity equals that of:
a) Circular motion
b) **Free fall from height** h
c) Sound in water
d) Light

Correct Answer: b

16. Energy lost in viscous flow is converted into:
a) Potential energy
b) Kinetic energy
c) **Heat (thermal energy)**

Correct Answer: c

17. Which device measures fluid speed using a manometer?
a) Thermometer
b) **Venturi meter**
c) Barometer
d) Hygrometer

Correct Answer: b

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18. The Magnus effect explains:

a) Plane stalling

b) **Swing of a cricket ball**

c) Manometer operation

d) Pipe flow

Correct Answer: b

19. On a spinning ball, the side with higher airflow speed experiences:

a) **Lower pressure**

b) Higher pressure

c) No pressure change

d) Double gravity

Correct Answer: a

20. The side with increased pressure corresponds to:

a) High speed

b) **Low speed**

c) Zero flow

d) Infinite speed

Correct Answer: b

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Section 2: Viscosity, Artery Flow & Reynolds Number

21. Poiseuille's equation applies to:

a) Ideal fluids

b) **Viscous Newtonian fluids**

c) Turbulent air

d) Vacuum

Correct Answer: b

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22. Velocity profile in viscous pipe flow is:

a) Linear

b) Square

c) **Parabolic**

d) Exponential

Correct Answer: c

23. Velocity at the pipe wall is:

a) Maximum

b) **Zero**

c) Average

d) Half of center velocity

Correct Answer: b

24. Adhesive forces act between:

a) Same fluid molecules

b) **Fluid and surface**

c) Fluid and gravity

d) Two gases

Correct Answer: b

25. Reynolds number for artery flow is:

a) $\rho v/\eta$

b) $\rho vL/\eta$

c) vL/ρ

d) ρgh

Correct Answer: b

26. Laminar flow occurs when:

a) $Re > 4000$

b) $Re < 2000$

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c) $Re = 0$

d) Infinite

Correct Answer: b

27. A Reynolds number of 8.1 indicates:

a) Turbulent flow

b) Transitional

c) **Laminar flow**

d) Supersonic

Correct Answer: c

28. Viscosity of blood used was:

a) 1.06×10^{3}Pa·s

b) 2.084×10^{-3}**Pa·s**

c) 5.03×10^{-5}Pa·s

d) 1.00×10^{-6}Pa·s

Correct Answer: b

29. Average blood velocity was found using:

a) Bernoulli equation

b) **Continuity equation**

c) Newton's law

d) Torricelli's formula

Correct Answer: b

30. Pressure drop in arteries is caused by:

a) Speed

b) Gravity

c) **Viscosity**

d) Atmosphere

Correct Answer: c

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Section 3: Newton's Laws and Momentum

31. Newton's Second Law relates force to change in:
    a) Position
    b) **Linear momentum**
    c) Temperature
    d) Volume

Correct Answer: b

32. Linear momentum equals:
    a) Mass × acceleration
    b) **Mass × velocity**
    c) Force × area
    d) Pressure × volume

Correct Answer: b

33. Momentum is conserved when:
    a) Gravity exists
    b) **Net force is zero**
    c) Speed constant
    d) Mass zero

Correct Answer: b

34. Angular form $M = I\alpha$ relates torque to:
    a) Linear velocity
    b) **Angular acceleration**
    c) Mass
    d) Density

Correct Answer: b

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35. Moment of inertia measures resistance to:

a) Translation

b) **Rotation**

c) Volume change

d) Area

Correct Answer: b

36. Angular momentum is conserved when:

a) Net force is zero

b) **Net torque is zero**

c) Velocity constant

d) Body is at rest

Correct Answer: b

37. A control volume is:

a) Fixed tank only

b) **Any region in space**

c) Zero pressure region

d) Earth

Correct Answer: b

38. A control volume with changing shape is:

a) Fixed

b) Moving

c) **Deforming**

d) Static

Correct Answer: c

39. Forces on a control volume are:

a) Internal & external

b) **Body & surface forces**

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c) Primary & secondary
d) Scalar & vector

Correct Answer: b

40. Most common body force is:
a) Pressure
b) Friction
c) **Gravity**
d) Magnetism

Correct Answer: c

Section 3 (Continued): Newton's Laws and Momentum

41. Surface forces include:
a) Gravity only
b) **Pressure and viscous forces**
c) Electric forces
d) Reaction forces only

Correct Answer: b

42. Normal stresses on a surface are composed of:
a) Gravity and friction
b) **Pressure and viscous stresses**
c) Only shear stress
d) Atmospheric pressure only

Correct Answer: b

43. Shear stresses are composed entirely of:
a) Pressure
b) **Viscous stresses**
c) Gravity
d) Weight

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Correct Answer: b

44. Atmospheric pressure is often ignored in force balances because:
a) It is too small
b) **It cancels out in all directions**
c) It acts downward only
d) It exists only in vacuum

Correct Answer: b

45. The Linear Momentum Equation is primarily used to calculate:
a) Fluid temperature
b) **Forces induced by flow**
c) Fluid density
d) Atomic mass

Correct Answer: b

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46. An inertial coordinate system:
a) Rotates rapidly
b) **Moves at constant velocity or is at rest**
c) Accelerates continuously
d) Is at Earth's center

Correct Answer: b

47. Reynolds Transport Theorem for momentum is used to derive:
a) Bernoulli equation
b) **Linear Momentum Equation**
c) Continuity equation
d) Poiseuille's law

Correct Answer: b

48. Rotational motion occurs when all points move in circles about:
a) Center of mass only

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b) **An axis of rotation**
c) The ground
d) A moving point

Correct Answer: b

49. The moment arm is the perpendicular distance from the axis to the:
a) Origin
b) **Line of action of force**
c) Center of body
d) Top of body

Correct Answer: b

50. The Angular Momentum Equation is also known as the:
a) Energy equation
b) **Moment of momentum equation**
c) Force equation
d) Radial equation

Correct Answer: b

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Section 4: Turbo Machines and Angular Momentum

51. Which devices are analyzed using angular momentum equations?
a) Stationary pipes
b) **Turbo machines**
c) Manometers
d) Sluice gates

Correct Answer: b

52. Moment of a force is the vector product of:
a) Mass and velocity
b) **Position vector and force**

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c) Torque and time
d) Area and pressure

Correct Answer: b

53. A rotating lawn sprinkler demonstrates:
a) Linear momentum
b) **Angular momentum**
c) Torricelli's formula
d) Archimedes' principle

Correct Answer: b

54. In steady flow, net torque equals the difference between:
a) Pressure and gravity
b) **Outgoing and incoming angular momentum**
c) Force and area
d) Density and viscosity

Correct Answer: b

55. Radial-flow devices involve flow:
a) Parallel to axis
b) **Normal to axis**
c) Only at inlet
d) In vacuum

Correct Answer: b

56. Axial-flow devices are analyzed using:
a) Angular momentum
b) **Linear momentum**
c) Energy equation
d) Ideal gas law

Correct Answer: b

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57. Moment direction is determined by:
a) Left-hand rule
b) **Right-hand rule**
c) Thumb rule
d) Gravity

Correct Answer: b

58. Without external moments, angular momentum:
a) Increases
b) **Is conserved**
c) Decreases
d) Becomes linear momentum

Correct Answer: b

59. Centrifugal pumps are:
a) Linear-flow devices
b) **Radial-flow devices**
c) Static devices
d) Thermal devices

Correct Answer: b

60. If moment of inertia is constant, net torque equals:
a) ma
b) $I\alpha$
c) PV
d) mgh

Correct Answer: b

Section 5: Dimensional Analysis

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61. Very few real fluid problems can be solved by:
a) Experiments
b) **Analytical analysis alone**
c) Computer models
d) Observation

Correct Answer: b

62. Concept allowing lab results to apply to real systems is:
a) Gravity
b) **Similitude**
c) Relativity
d) Entropy

Correct Answer: b

63. A laboratory system is called a:
a) Prototype
b) **Model**
c) Template
d) Replica

Correct Answer: b

64. A dimension is a measure of quantity:
a) With numbers
b) **Without numerical values**
c) Only meters
d) Only kilograms

Correct Answer: b

65. A unit is:
a) A physical quantity
b) **A way to assign number to dimension**

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c) Same as dimension
d) Mathematical constant

Correct Answer: b

66. Number of primary dimensions:
a) Three
b) Five
c) **Seven**
d) Ten

Correct Answer: c

67. Which is NOT a primary dimension?
a) Mass
b) Length
c) **Velocity**
d) Temperature

Correct Answer: c

68. Dimensional homogeneity requires all terms to have:
a) Value one
b) **Same dimensions**
c) Gravity
d) Primary units

Correct Answer: b

69. A classic example of dimensional homogeneity is:
a) Einstein's equation
b) **Bernoulli equation**
c) Poiseuille equation
d) Torricelli formula

Correct Answer: b

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70. Dimensions of pressure are:
a) L/t
b) $m/(Lt^2)$
c) mL/t^2
d) m/L^3

Correct Answer: b

Section 6: Similarity and Pi Theorem

71. A normalized nondimensional equation has terms of order:
a) Zero
b) **One**
c) Infinity
d) Ten

Correct Answer: b

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72. Nondimensional numbers are often named after:
a) Planets
b) **Scientists**
c) Colors
d) Animals

Correct Answer: b

73. Inspectional analysis means:
a) Experiments
b) **Nondimensionalizing equations**
c) Observation
d) Calibration

Correct Answer: b

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74. In nondimensional Bernoulli equation, terms are:
a) Pressure
b) Energy
c) **Dimensionless**
d) Length

Correct Answer: c

75. Dimensional variables are those that:
a) Never change
b) **Vary in the problem**
c) Are dimensionless
d) Are always primary

Correct Answer: b

76. A pure constant is:
a) Gravity
b) **π**
c) Density
d) Length

Correct Answer: b

77. Scaling parameters usually include:
a) Color & taste
b) **Characteristic length and velocity**
c) Temperature only
d) Mass only

Correct Answer: b

78. Froude number is important in:
a) Space
b) **Free-surface flows**

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c) Arteries
d) Gas expansion

Correct Answer: b

79. Nondimensionalization yields parameters like:
a) **Reynolds & Froude numbers**
b) Mass
c) Time
d) Area

Correct Answer: a

80. Model testing saves:
a) **Time and money**
b) Accuracy
c) Dimensions
d) Pressure

Correct Answer: a

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Section 7: Advanced & Miscellaneous Concepts

81. Complete similarity requires:
a) Geometric
b) Kinematic
c) Dynamic
d) **All of the above**

Correct Answer: d

82. Geometric similarity means:
a) Same size
b) **Same shape**

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c) Same weight
d) Same speed

Correct Answer: b

83. Kinematic similarity requires velocities to be:
a) Equal
b) **Proportional**
c) Zero
d) Opposite

Correct Answer: b

84. Dynamic similarity requires forces to:
a) Match lengths
b) Match velocities
c) **Scale proportionally**
d) Match time

Correct Answer: c

85. Pi (Π) denotes:
a) Primary dimension
b) **Dimensionless parameter**
c) Scaling law
d) Pressure

Correct Answer: b

86. Π_1 is usually the:
a) Independent parameter
b) **Dependent parameter**
c) Constant
d) Variable

Correct Answer: b

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87. For complete similarity, all ____ Pi groups must match:

a) Dependent

b) **Independent**

c) Primary

d) Secondary

Correct Answer: b

88. Reynolds number equals ratio of:

a) **Inertial to viscous forces**

b) Pressure to gravity

c) Mass to volume

d) Velocity to sound

Correct Answer: a

89. Most common dimensionless number is:

a) Froude

b) Mach

c) **Reynolds**

d) Weber

Correct Answer: c

90. Drag balance measures:

a) Temperature

b) **Aerodynamic drag**

c) Viscosity

d) Atomic mass

Correct Answer: b

Final Set

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91. Buckingham Pi theorem gives Pi groups as:
a) $n + r$
b) $n \times r$
c) $n - r$
d) r/n

Correct Answer: c

92. Important dimensionless groups include:
a) Mach
b) Weber
c) Euler
d) **All of the above**

Correct Answer: d

93. Dimensional analysis helps by:
a) Increasing variables
b) **Creating nondimensional groups**
c) Ignoring properties
d) Shrinking models

Correct Answer: b

94. Froude number dominates in:
a) Arteries
b) **Ship towing tank tests**
c) Gas turbines
d) Vacuum

Correct Answer: b

95. Inspectional analysis is applied to:
a) Raw data
b) **Equations of motion**

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c) Equipment
d) Humans

Correct Answer: b

96. Nondimensionalization requires selecting:
a) Random numbers
b) **Scaling parameters**
c) New coordinates
d) Fluid color

Correct Answer: b

97. Weber number relates to:
a) Gravity
b) **Surface tension**
c) Viscosity
d) Sound

Correct Answer: b

98. A prototype is the:
a) Lab model
b) **Full-scale system**
c) Formula
d) Scale factor

Correct Answer: b

99. Similarity allows development of:
a) Dimensions
b) **Predictions and correlations**
c) New fluids
d) Variables

Correct Answer: b

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100. Strouhal number appears in:
a) Steady flow
b) **Unsteady flow**
c) Ideal flow
d) Static fluids

Correct Answer: b

Section 7 (Continued): Specific Concepts and Formulae

101. The equation $A_1 v_1 = A_2 v_2$ is known as the:
a) Bernoulli Equation
b) **Equation of Continuity**
c) Linear Momentum Equation
d) Navier–Stokes Equation

Correct Answer: b

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102. If $A_1 > A_2$, then according to continuity:
a) $v_1 > v_2$
b) $v_1 < v_2$
c) $v_1 = v_2$
d) $v_1 = 0$

Correct Answer: b

103. In the Venturi effect, when area decreases:
a) **Velocity increases and pressure decreases**
b) Velocity decreases and pressure increases
c) Both increase
d) Both decrease

Correct Answer: a

104. Internal energy of a fluid increases in viscous flow due to:
a) Increased velocity

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b) **Heating due to friction**
c) Gravity
d) Potential energy

Correct Answer: b

105. Viscous stresses in a fluid include:
a) Only normal stress
b) **Normal and shear stresses**
c) Only atmospheric stress
d) No stresses

Correct Answer: b

106. Pressure P in Bernoulli's equation is also called:
a) Kinetic pressure
b) **Static pressure (pressure energy density)**
c) Total pressure
d) External pressure

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Correct Answer: b

107. Moment of inertia of a point mass m at distance r is:
a) mr
b) mr^2
c) m/r
d) r/m

Correct Answer: b

108. Torque is the rate of change of:
a) Linear momentum
b) **Angular momentum**
c) Kinetic energy
d) Mass

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Correct Answer: b

109. Flow in a centrifugal pump is in the:
a) Axial direction
b) **Radial direction**
c) Circular direction
d) Tangential direction

Correct Answer: b

110. First step of Buckingham Pi theorem is to list:
a) Primary dimensions
b) **All dimensional variables (n)**
c) Constants
d) Final answer

Correct Answer: b

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Section 8: Miscellaneous and Deeper Insights

111. Arteriosclerosis causes vascular flutter because:
a) Pressure increases in constrictions
b) **Low internal pressure allows artery collapse**
c) Blood stops flowing
d) Heart beats faster

Correct Answer: b

112. Dimensions of velocity are:
a) mL/t
b) L/t
c) t/L
d) L^2/t

Correct Answer: b

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113. Dimensions of force are:
a) m/t^2
b) mL/t^2
c) mL/t
d) L/t^2

Correct Answer: b

114. The dimension representing "amount of matter" is:
a) Mass
b) **Mole**
c) Temperature
d) Length

Correct Answer: b

115. A flow with no external moments obeys:
a) Linear momentum conservation
b) **Angular momentum conservation**
c) Bernoulli principle
d) Poiseuille's law

Correct Answer: b

116. In the Magnus effect, the ball moves toward the side of:
a) Higher pressure
b) **Lower pressure**
c) Higher gravity
d) Higher viscosity

Correct Answer: b

117. SI unit of viscosity is:
a) Pa
b) **Pa·s**

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c) N/m
d) kg/m

Correct Answer: b

118. SI unit of volume flow rate $Q$is:
a) m^2/s
b) **m^3/s**
c) kg/s
d) m/s

Correct Answer: b

119. Cavitation number is related to:
a) Sound waves
b) **Vapor pressure and cavitation**
c) Surface tension
d) Elasticity

Correct Answer: b

120. A well-rounded pipe entrance justifies assuming:
a) Turbulent flow
b) **Uniform flow approximation**
c) Zero flow
d) Fully viscous flow

Correct Answer: b

Final Questions

121. Reynolds Transport Theorem connects:
a) Atoms and molecules
b) **System and control volume analysis**

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c) Pressure and temperature
d) Models and prototypes

Correct Answer: b

122. Linear velocity V relates to angular velocity ω as:
a) $V = \omega/r$
b) $V = r\omega$
c) $V = \omega^2 r$
d) $V = r/\omega$

Correct Answer: b

123. "Strength of rotational effect" refers to:
a) Power
b) **Torque (moment)**
c) Momentum
d) Inertia

Correct Answer: b

124. Mass flow rate is defined as:
a) ρAV
b) P/V
c) m/t
d) A/V

Correct Answer: a

125. Gauge pressure is calculated by:
a) Adding atmospheric pressure
b) **Subtracting atmospheric pressure**
c) Multiplying by gravity
d) Dividing by area

Correct Answer: b

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126. Viscous drag on a sphere depends on:
a) Two parameters
b) **Four parameters (D, V, ρ, μ)**
c) One parameter
d) Six parameters

Correct Answer: b

127. If $n = 5$and $r = 3$, number of Pi groups is:
a) 8
b) 15
c) **2**
d) 5

Correct Answer: c

128. In artery Reynolds number, $L = 2R$represents:
a) Radius
b) **Diameter**
c) Circumference
d) Area

Correct Answer: b

129. Bernoulli's principle explains why air speed on top of wing is:
a) Lower
b) **Higher**
c) Zero
d) Constant

Correct Answer: b

130. Energy density in Bernoulli's equation means energy per unit:
a) Mass
b) **Volume**

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c) Area
d) Time

Correct Answer: b

131. Static pressure is represented by:
a) $\frac{1}{2}\rho v^2$
b) P
c) ρgh
d) ma

Correct Answer: b

132. Curving of a cricket ball demonstrates:
a) Uniform motion
b) **Magnus effect**
c) Static equilibrium
d) Linear flow

Correct Answer: b

133. A deforming control volume is one where:
a) Whole volume moves
b) **Part of boundary moves relative to others**
c) It is fixed
d) It is circular

Correct Answer: b

134. Primary dimension for temperature is denoted by:
a) T
b) **Θ (Theta)**
c) K
d) C

Correct Answer: b

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135. Pressure drag is caused by:
a) Skin friction
b) **Flow separation and wake formation**
c) Altitude
d) Vacuum

Correct Answer: b

136. Geometric similarity scale factor is:
a) Π
b) L_p/L_m
c) Reynolds number
d) Gravity

Correct Answer: b

137. Kinematic viscosity is defined as:
a) $\mu\rho$
b) μ/ρ
c) ρ/μ
d) $1/\mu$

Correct Answer: b

138. Total force on a control volume includes:
a) Body forces only
b) **Body and surface forces**
c) Gravity only
d) Pressure only

Correct Answer: b

139. A faucet assembly highlights the importance of:
a) **Choosing a proper control volume**
b) Large pipes

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c) Turning off water
d) High pressure

Correct Answer: a

140. Subsonic speeds are:
a) Faster than light
b) **Slower than sound**
c) Exactly sonic
d) In vacuum

Correct Answer: b

141. The theorem converting system analysis to control volume is:
a) Pythagoras theorem
b) **Reynolds Transport Theorem**
c) Buckingham theorem
d) Mean value theorem

Correct Answer: b

142. Direction of torque is:
a) Parallel to r
b) **Perpendicular to plane of $r$and F**
c) Toward origin
d) Opposite force

Correct Answer: b

143. Angular momentum equation helps aircraft determine:
a) Fuel level
b) **Angular acceleration when thrust is applied**
c) Heart rate
d) Radio signal

Correct Answer: b

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144. Length of the 1:46.6 scale destroyer model was:
a) 100 m
b) **3.048 m**
c) 46.6 m
d) 1 m

Correct Answer: b

145. Primary dimensions are also called:
a) Derived
b) **Fundamental dimensions**
c) Non-primary
d) Complex

Correct Answer: b

146. Experimental data is required because very few problems of _____ can be solved analytically:
a) Ideal fluids
b) **Real fluids**
c) Static fluids
d) Air

Correct Answer: b

147. Newton's First Law states motion remains unchanged if net force is:
a) Positive
b) **Zero**
c) Negative
d) Maximum

Correct Answer: b

148. Momentum flux correction factor is denoted by:
a) α
b) **β**

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c) γ
d) δ

Correct Answer: b

149. Dynamic pressure is defined as:
a) $\rho g h$
b) $\frac{1}{2}\rho v^2$
c) P
d) ma

Correct Answer: b

150. Boundary layer thickness depends mainly on:
a) Temperature
b) **Reynolds number**
c) Color of fluid
d) Pipe length only

Correct Answer: b

151. As Reynolds number increases, boundary layer becomes:
a) Thicker
b) **Thinner**
c) Constant
d) Zero

Correct Answer: b

152. Turbulent flow occurs at:
a) Low Reynolds number
b) **High Reynolds number**
c) Zero velocity
d) Absolute zero

Correct Answer: b

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153. Momentum flux correction factor accounts for:

a) Pressure loss

b) **Non-uniform velocity profile**

c) Temperature change

d) Gravity effect

Correct Answer: b

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